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Current status and future trends of offshore wind power in Europe

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ABSTRACT

A considerable growth of large-scale offshore wind farms (OWF) is noticeable in Europe mainly due to advances in wind turbines and foundation structures, which have improved their economic conditions and contributed to the implementation of offshore plants. It is expected that the installed capacity will continue to increase, since the European Union aims at reaching about 100 GW of offshore wind capacity by 2030. OWF have many challenges in terms of cost reduction and technology. Due to these issues, it is important to have a comprehensive understanding of the current situation and the future trends of OWF. This paper analyses the current status and the trends of large-scale OWF with an installed capacity above 150 MW in Europe. The results show the trends on wind turbine size and capacity, turbine model, distance to shore, water depth, investment cost, type of foundation, transmission technology, and voltage array systems among others. This paper gathers the latest information about the topic, deducing future trends from the evaluation of OWF fully commissioned, authorized or under construction.
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1. Introduction

Offshore wind farms (OWF) show favorable indicators in almost every environmental impact index investigated, which is aligned with the search of sustainability in electric power generation [1]. OWF share all the benefits of onshore power plants and present further advantages. For instance, larger areas available for bigger plants, greater plants in terms of installed capacity, steadier wind speeds with less turbulence, and lower visual and acoustic impacts [2].

However, these systems also present certain drawbacks, such as limited access to the sea, costly marine foundations, and connection to the grid through marine electric lines. Those disadvantages are attenuated by the possibility to have a higher capacity factor in OWF [2–4]. Moreover, to offset the higher costs associated with OWF, the developers are focusing on increasing the installed capacity [5].

In this sense, an increasing trend towards large-scale based systems has been reported in the literature [3,6], although the fast growth experienced after 2013 was not fully addressed. The perspective and challenges in the development of offshore wind power were highlighted in Ref. [3], mainly dealing with the potential interests of this sector, although the authors did not differentiate by country in their analysis as in Ref. [6], where the authors also explored some technical, economic and environmental issues.

The rearrangement in OWF has been significant since 2013 in terms of types and models of turbines, transmission systems, increase of the installed capacity and distance to shore. Rodrigues et al. brought out information of interest related to the topic, although not emphasizing on large-scale power plants [7]. The commitment towards the reduction of greenhouse emissions from electricity generation is reinforced by the importance of wind in the energy mix [8]. Some reviews about OWF have been performed separately by country. For instance, an inquiry into offshore in the United Kingdom (UK) highlighted an increase and accelerated development of offshore wind energy in this country [9], whereas [10] presented an overview of the status and future development in

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Spain in 2016.

Regarding the wind turbines, there are different technologies available for OWF. In this sense, doubly-fed induction generator (DFIG), squirrel cage induction generator (SCIG) with full converter and permanent magnet synchronous generator (PMSG) are usually chosen [11–13]. Arrambide et al. analyzed the wind turbine technology used in OWF above 25 MW in Ref. [14]. The authors highlighted the challenging seek for the appropriate technology, but future trends were not evaluated for large-scale OWF in Europe.

In terms of the power transmission technologies, the preference for a high voltage alternate current (HVAC) transmission against direct current was indicated in Ref. [15] for OWF located up to 50 km away from the shore to avoid expensive converter stations and increased power losses. For OWF far from the shore, high voltage direct current (HVDC) transmission is more suitable. Additionally, long HVAC transmission lines need reactive compensation to be technically viable, while HVDC does not show such limitation [16]. On the negative side, costs in HVDC lines are typically higher than in HVAC [17]. Another technology, the low-frequency AC (LFAC) transmission, was presented as a potential competitor to HVDC for medium/long-distance (50–200 km from the shore) OWF in Ref. [18]. A thorough comparison of HVAC and HVDC, as well as the combination of both, was carried out in Ref. [19]. Nevertheless, both [18,19] skip the economic analysis of OWF in operation in Europe.

The foundation structures of offshore wind turbines are crucial elements in the development of OWF. Different types of foundations (i.e. monopile, jacket, tripod, gravity, and floating) are recommended depending on the depth of the power plant, as discussed in Ref. [20], where the future challenges for structures and the research of soil dynamics were presented. Furthermore, various types of structures and examples of OWF projects were shown in Ref. [21], although not all the existing OWF in Europe were included in the study.

The economy of the OWF project is associated with its viability. In this sense, it is essential to contrast their profitability against onshore [2,22]. Some important factors in the economic evaluation of OWF are the acquisition/installation of wind turbines and foundations, the electrical infrastructure, the design and project management, and the operation and maintenance [23]. Some economic criteria related to the pre-investment and investment costs for OWF equipped with AC transmission were analyzed in Ref. [24]. Although OWF require large investments, a gradual fall is foreseen from the mid-2020s [25].

To understand the development of OWF, it is very important to identify the trends that are driving the sector in certain vital aspects. In general, most of the information available is scattered around extensive reports updated yearly [26–31], while other very specific information can only be found in individual project documentation [32–35]. This makes the perception of trends very complicated and laborious. To the best of our knowledge, updated articles that extensively analyze large-scale OWF in Europe cannot be found. In this work, the latest information demonstrating the status and trends of existing OWF above 150 MW in Europe is gathered and analyzed. The trends are derived and justified with actual data obtained from official reports and literature, not only based on the impressions of the authors or other experts. The primary aim of this study is to establish the future trends in OWF and demonstrate the wide growth of this technology, especially in large-scale OWF, where the sector is very economically competitive. With this purpose, the most significant characteristics of 57 plants installed in Europe with a rated power above 150 MW and fully commissioned until 2019, as well as 11 plants authorized or under construction, are studied in detail, drawing relevant conclusions from the data collected. Significant evidences of large-scale OWF in operation were not found in other locations except for China, which is a market in constant development. The main contribution of this work is to provide specialists in the field with a significant amount of data gathered from several technical reports, which have been used to validate and quantify the future evolution of OWF. Additionally, this article might also be pertinent to researchers approaching this topic for the first time, as they will access a large amount of information summarized in a single document, and will get to learn some relevant trends that might not be obvious to the inexpert eye.

2. Methodology

The main objective of this paper is to study and analyze OWF currently in operation or projected in Europe, paying special attention to plants above 150 MW, justified by the fact that large power plants above this reference value are becoming increasingly common in offshore applications.

Initially, the characteristics of the OWF studied are discussed in terms of their installed capacity, size, distance to shore and water depth. Next, the status and trends of large offshore in Europe are revised. All the results were extracted from plants in operation or authorized for construction, and some of the results presented were distributed by country.

The necessary information was collected from the available literature, mainly using specialized databases in Ref. [36] and reports from governments and manufacturers [26,28–30,37–42]. Based on this, a database was implemented with 57 plants in operation and 11 plants authorized or under construction, all of them with a rated power above 150 MW and located in Europe.

The data collected include varied information used for an exhaustive analysis to establish standards for the data, to reduce the uncertainties of the results, and to create a document of quick access that shows the tendencies of OWF.

On the first stage of the analysis, standards of depth and distance to the coast over the years are set according to the data. The distance to the coast was computed from the center of the OWF. Regarding the depth of the foundations, the values declared by the developer were considered. In this sense, the average depth of the plants was calculated as the mean value between the deepest and the shallowest turbine structure. In addition, the installed capacity and the number of turbines were also observed.

On a second stage, the data were classified by country, turbine technology, type of transmission and operating or under construction plants, and subjected to a more exhaustive analysis. This methodology made it possible to extract information and trends for those characteristics, mostly using regression techniques from previous data. When regression was not used, the alternative technique employed is stated in the text.

When different monetary values and years were mixed in the analysis, a monetary normalization was carried out as described in Ref. [43]. All values were normalized to Euros (€) or Euros/MW and referenced to 2019, in order to diminish the uncertainties in the cost analysis. In this regard, the data were first corrected before the inflation and later converted to the respective value in Euros [43].

3. Current status and trends of large OWF in Europe

This section explores general trends in OWF in Europe and presents the current situation. In a first approach, OWF all over Europe have been observed, regardless of their rated power. After that, the study focuses on OWF above 150 MW.

3.1. Europe's OWF

In Europe, OWF will continue to grow and increase their contribution to the energy mix. This trend is favored by the support that renewable energies receive from regulatory policies, fiscal incentives and public funding, which increase their competitiveness [44].

Eleven European countries contributed to the installed capacity of OWF until 2019, where the UK and Germany stand out with the largest capacity. The offshore technologies are growing in Europe, as seen in Table 1, which shows the evolution of offshore energy over the years, as well the expected trends for 2020, 2025 and 2030. The data presented in Table 1 were obtained from Refs. [26,28,30,37,40,45–49].

The trends for 2020, 2025 and 2030 were estimated from the number and average power of OWF in the period between 2008 and 2019 using linear regression. For these estimations, a neutral scenario was taken into account, which considers that the investments on OWF will remain the same when compared to the current scenario. Thus, the number of OWF (N_{WF}) can be deduced from (1), whereas the average power of wind farms (A_{PF}) is given by (2), obtained from Fig. 1, where x indicates the year of projection. These trends are presented in Table 1.

$$N_{WF} = 6.73 \cdot x - 13489.13 \tag{1}$$

$$A_{PF} = 46.1 \cdot x - 92483.86 \tag{2}$$

The 25.4 GW of installed capacity for 2020 were calculated using the projected average growth of 3.3 GW for 2020. Therefore, the installed capacity calculated for 2020 is the sum of the 22.07 GW installed in 2019 plus 3.3 GW. The values of 47.4 and 76 GW included in Table 1 correspond to a neutral scenario for 2025 and 2030, i.e.: considering the same growth rate as in the previous years. These values were found through the polynomial regression presented in (3), deduced from Fig. 2, where IC indicates the installed capacity.

$$IC = 140.74 \cdot x^2 - 564911.49 \cdot x + 566878201 \tag{3}$$

Finally, the number of wind turbines (N_{WT}), was estimated using the polynomial regression in (4), obtained from Fig. 2.

$$N_{WT} = 24.97 \cdot x^2 - 100092.35 \cdot x + 100321939.49 \tag{4}$$

From Table 1, it can be observed that, under the scenarios considered, the generation capacity increases more sharply than the number of generators installed, indicating that the rated power of the wind turbines installed in OWF shows a tendency to grow.

The results estimated through linear and polynomial regression were close to the values found in Refs. [50,51], which describe possible scenarios for 2030. In Refs. [50,51], three scenarios for the installed capacity in 2030 were evaluated. In the first report, a neutral scenario of 66 GW, an optimistic projection with values that reach 98 GW, and a pessimistic projection of 44.6 GW were considered; while the second report indicates a neutral scenario of 70 GW, an optimistic projection of 99 GW, and a pessimistic projection of 49 GW. This trend towards a growing rated power of the wind turbines in European OWF is also highlighted in Refs. [26,28,29].

3.2. Europe's large-scale OWF

Until 2019, about 89% of the total installed capacity in European OWF corresponds to plants with rated power above 150 MW. For this reason, this section focuses on large-scale OWF.

The parameters evaluated for large-scale OWF are depicted in Fig. 3. When applicable, the data are differentiated by the color, size and line style of the circle to appreciate the variation taking 2013 as the reference year, when OWF experienced a fast growth in many segments. The same applies to the cluster diagrams in the subsequent sections. Nevertheless, some parameters do not have a red circle in Fig. 3. This is justified because some of them can be further sub-divided and they are dealt in the corresponding sub-sections; or simply because they cannot be quantified, as it is the case of the array systems. Additionally, the cluster diagrams of some subsections also indicate numerically the installed power belonging to the cluster, the percentage of the cluster installed power compared to the total installed power, and the average power of the wind turbines included in the cluster.

3.2.1. Installed capacity and size

Only in 2019, an increase of nearly 3412 MW new installed capacity was seen, with around 504 new turbines connected in Europe [26]. The trends indicate that these figures will continue to increase in the coming years.

Currently, Europe is the leader in offshore wind power generation and experiences a fast growth. The total installed capacity in the years 2008 and 2019 was approximately 1.5 and 22 GW respectively [26,40]. Regarding exclusively plants over 150 MW, they contribute with 89% of the total. Fig. 4 illustrates the increase on the installed capacity of OWF above 150 MW between 2002 and 2019. From 2013 onwards, about 76% of the new plants installed are above 150 MW.

Another aspect worth noting is that about 98% of OWF installed capacity in Europe is concentrated in only five countries. Considering plants above 150 MW, the UK and Germany sum 80% of the installed capacity, and the 20% remaining is divided among Belgium, Denmark and the Netherlands, with 8%, 8% and 4% of the installed capacity respectively. The UK represents 44% of the offshore wind turbines installed and in operation, therefore being the country with the largest installed capacity. For more details, a thorough review of the offshore generation in the UK is presented in Ref. [9].

3.2.2. Distance to shore and water depth

Most of the area available for OWF is located far from the coast (above 50 km). In this sense, Fig. 5 summarizes the status of OWF in terms of distance to shore and water depth. The black circles contain the total installed power until 2013, its percentage

Table 1
Growth of OWF in Europe over the years.

Current average													Trends		
Years	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2025	2030
Number of OWF	30	48	45	53	55	69	74	80	81	92	105	110	112	139	172
Number of Turbines	629	828	1136	1371	1662	2080	2488	3230	3589	4149	4543	5047	6651	11126	16850
Installed Capacity (GW)	1.48	2.06	2.95	3.81	4.99	6.56	8.05	11.03	12.63	15.78	18.5	22.07	25.3	47.4	76
Average Power (MW)	62.2	72.2	155.3	200	285.6	485	386	337.9	379.5	493	561	621	630	849	1080

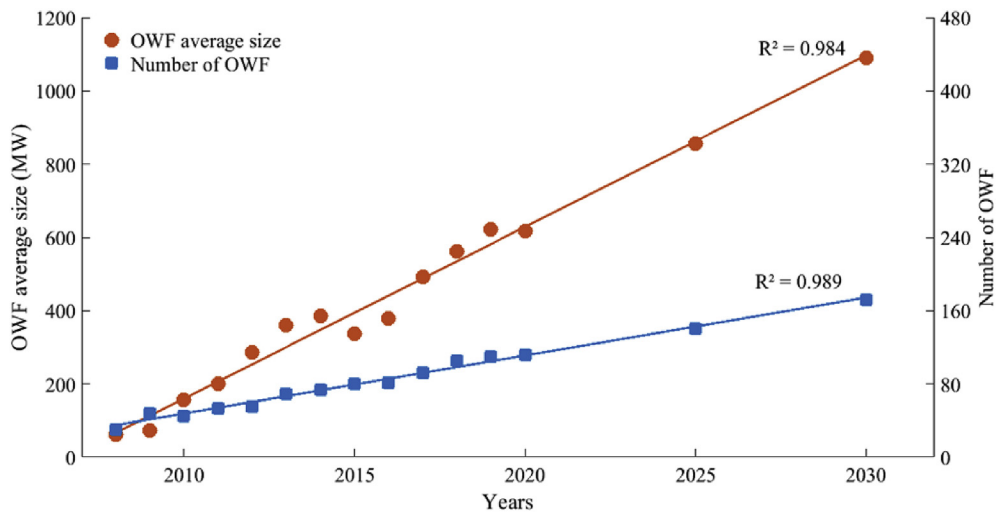

Fig. 1. Average power and number of OWF in Europe.

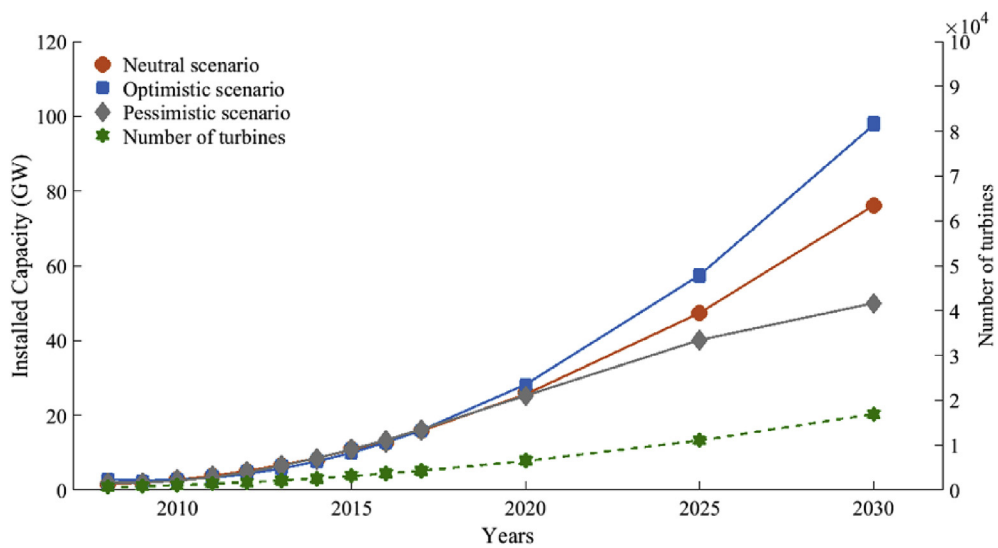

Fig. 2. Trends in installed capacity and number of offshore wind turbines in Europe.

compared to the total, and the average rated power of the offshore wind turbines included in each range. The red dashed circles contain the same parameters as the black circles from 2013 to 2019. For instance, in Fig. 5(a), the amount of offshore wind power installed for a distance to shore below 20 km was 4.28 GW, corresponding to 1.89 GW before 2013, and 2.39 GW after (and including) 2013. Regarding the percentage of the total installed power, these values correspond to 9.65% and 12.22%, before and after 2013 respectively, for a total installed capacity of 19.58 GW. Before 2013, no OWF with a distance to shore above 50 km had been reported, and for this reason, only the red circle is represented.

The distance to the shore is increasing over the years, and consequently, the wind turbines have to be installed in deeper waters. This trend can be observed in Fig. 6, which represents in separate coordinate axis the average depth and distance to shore of OWF from 2008 to 2019. Additionally, the average rated power of the turbines and OWF is indicated by the size of the blue and orange circles, respectively, together with the corresponding numerical values.

Fig. 6 shows a 193% increase between the deepest and shallowest OWF. Furthermore, the average distance in 2019 is approximately 4.5 times higher than in 2008, and the average power of the wind turbines is about 3.25 times larger for the same period. The average distance to shore and depth of OWF have increased in 132% and 60% respectively, since 2013. Regarding the deepest offshore wind generator, it is installed at about 56 m, and the furthest is less than 120 km away from the coast.

Fig. 7 shows the trends for the distance and depth for the years 2025 and 2030. A polynomial regression shows that the predicted depth for OWF above 150 MW for 2025 and 2030 is 44 and 53 m, respectively. When considering all OWF, these values decrease to 42 and 50 m, respectively. The distance predictions only for OWF above 150 MW are 80 and 100 km for 2025 and 2030, respectively. However, the average values for all the OWF decrease to 70 and 87 km, respectively.

3.2.3. *Turbine technologies*

Offshore wind turbines present certain differences compared to onshore wind turbines. They are designed to withstand severe

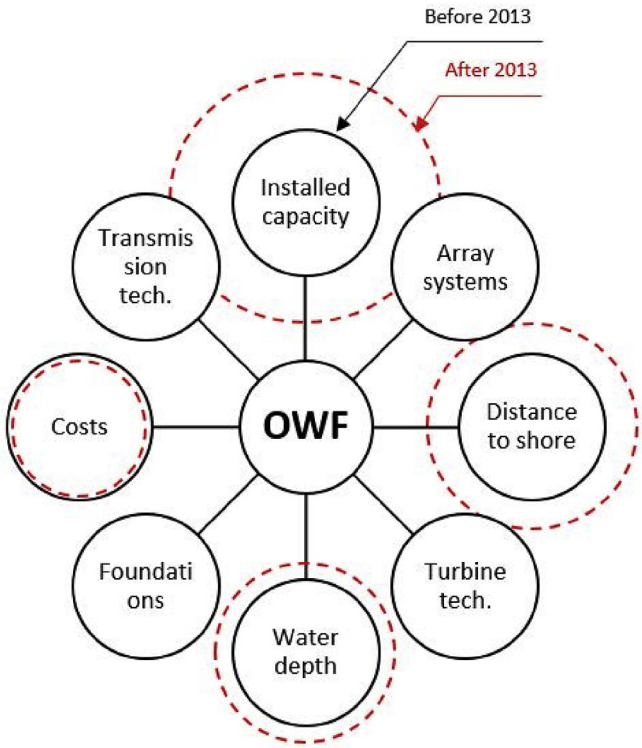

Fig. 3. Cluster diagram of the main parameters of OWF.

corrosion and stress conditions. Furthermore, offshore wind turbines can have larger rotor and blade sizes, as well as more powerful generators. Another important point is that the installation, operation, maintenance and electrical infrastructure costs decrease as the size of the offshore turbine increases. Therefore, the increase in the rated power of the offshore wind turbines has become a trend [6,52].

Among the different wind turbine technologies available, the DFIG, SCIG with full-scale converter and PMSG are usually employed in OWF, due to their higher energy efficiency [13]. Several generators used in OWF were presented in Ref. [14]. The relevance of each of these technologies is illustrated in Fig. 8. The offshore wind turbines are clustered into the aforementioned technologies. The total installed capacity until 2013, the percentage, and the average rated power of the turbines in each category are also indicated. The red dashed circles represent the variation in percentage of installed capacity from 2013 to 2019. It is possible to observe a modest increase in DFIG driven wind turbines in OWF commissioned after 2013, whereas the increase for SCIG is slightly higher. On the other hand, the PMSG technology started to be employed after 2013, and its installed capacity has increased significantly since then.

The generator models installed in Europe in large OWF between 2002 and 2019 can been found in Table 2. In total, 4307 wind turbines were installed and fully commissioned in European OWF above 150 MW during this period. The SCIG with full converter is currently the most used technology with 2180 turbines (51%), mainly due to its lower initial cost. The DFIG share corresponds to 562 turbines (13%), and they were widely used between 2009 and 2013. The PMSG reaches 1565 turbines (33%). However, it can be noticed that the trend for offshore turbines targets at this technology, since 1265 PMSG were installed between 2016 and 2019, against only 54 DFIG and 388 SCIG. Moreover, no DFIGs were installed in large wind farms in 2018 and 2019.

Another important trend for offshore turbines is the increasing concern of manufacturers with the reliability, durability and cost of these elements. In this sense, the industry leaders are developing modular offshore wind turbines. This concept consists in dividing the traditional single-pieced blades of the turbine into smaller pieces [52].

3.2.4. Size and number of turbines

The natural path of turbine manufacturers is the development of larger turbines in the next years. In general, when the power and size of the turbine increase, the costs of the foundations also increase. However, this increase in the rated power and size of the wind turbines reduces the number of units needed. Therefore, the total cost per MW tends to decrease with the growth of the rated power of turbines and plants.

Currently, offshore wind turbines have diameters ranging from 80 to 164 m and a rated power between 2 and 8.4 MW, with the largest turbines being manufactured by MHI Vestas [71] and

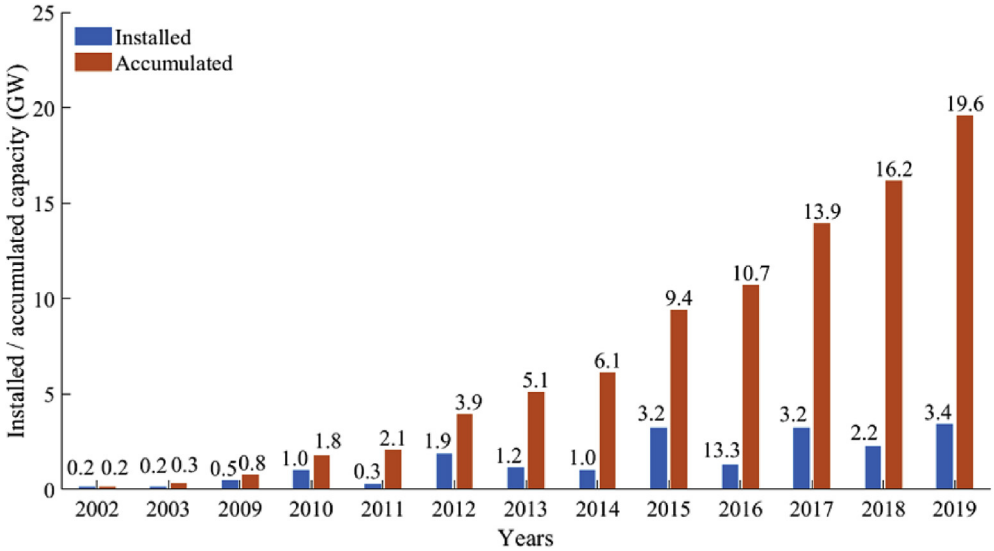

Fig. 4. Yearly installed and accumulated capacity of large-scale European OWF.

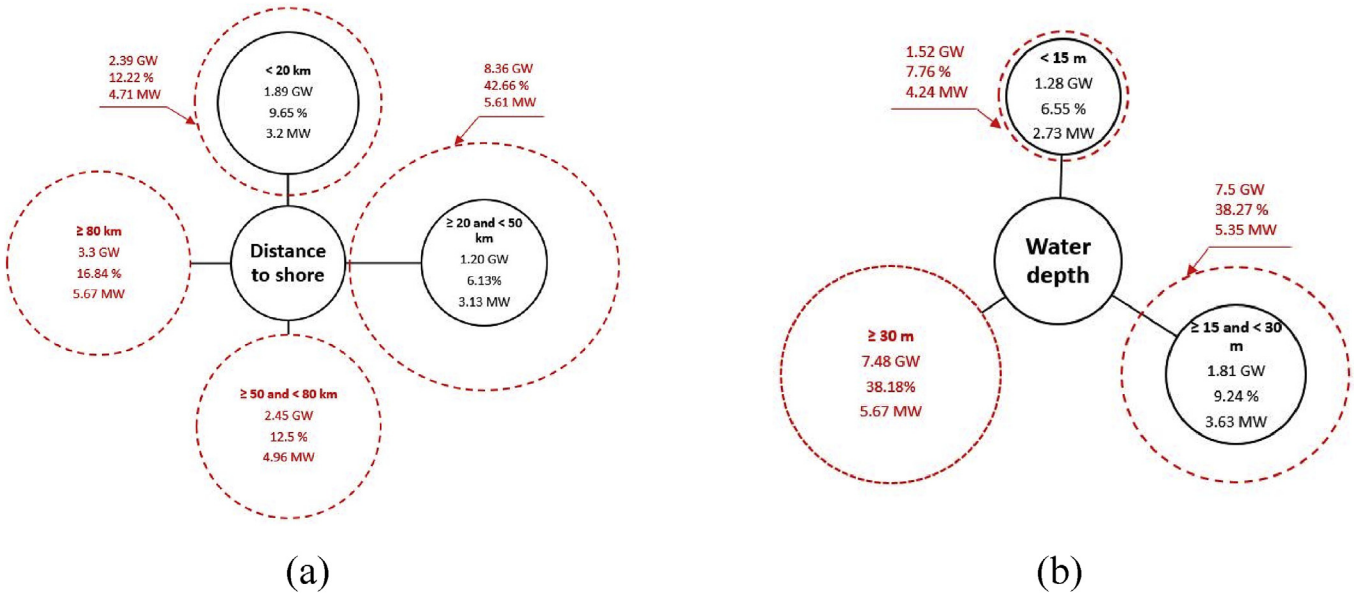

Fig. 5. (a) Distance to shore cluster diagram; and (b) Water depth cluster diagram.

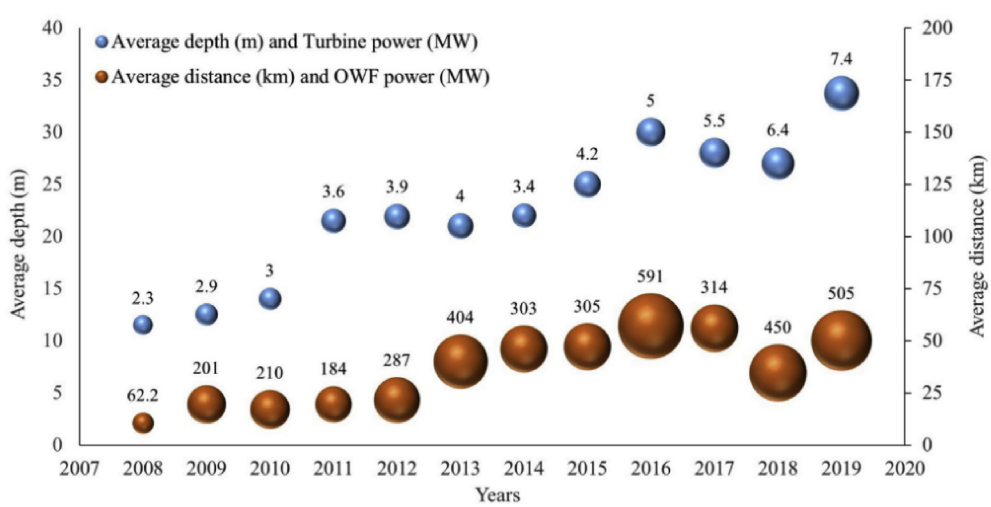

Fig. 6. Average depth and turbine power (blue) and average distance to shore and OWF power (orange). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Siemens Gamesa [72]. Hence, there is a growing tendency regarding the rated power and rotor diameter of turbines in OWF. The average size of the turbines in 2019 was 3.72 times bigger that in 2002. In Refs. [73,74], the feasibility of a 20 MW turbine was investigated, concluding that the project would be feasible and could significantly improve in the final cost of wind power plants, further revolutionizing this market.

Based on the information gathered, it can be indicated that, by 2025, generators ranging between 6 and 12.5 MW will be common in OWF, raising up to 20 MW by 2030 [74–76]. Similar values are obtained from the polynomial regression of Fig. 9. However, a linear regression with the values registered up to 2019 yields lower figures: 8.6 MW for 2025 and 10.7 MW for 2030. In Ref. [39], an average value of 8 MW for 2020 and 10 MW for 2030 is envisaged.

3.2.5. Transmission technologies

The trends for offshore transmission systems are presented in this section. Detailed studies of offshore transmission technologies can be found in Refs. [16,17,19,77–79].

In OWF, the transmission stage usually occurs at medium or high voltage AC (MVAC or HVAC), or high voltage DC (HVDC). Currently, MVAC transmission is an option only for plants near the coast (20 km maximum) and with an installed capacity up to 100 MW. This alternative is not profitable under other conditions according to Ref. [80]. Since 2002, HVAC has become the most used technology. This technology is consolidated, simple and more suitable for OWF with distances between 15 and 50 km from the shore. However, with an increase in the transmission distances, some drawbacks of the AC technology become more visible, such as a large amount of reactive current due to the high capacitance of the transmission lines, the unfeasible direct connection of two AC networks with different frequencies, and the failures in the HVAC cables. Based on these drawbacks, HVDC has become a technically viable and economical solution in cases of large distance and power

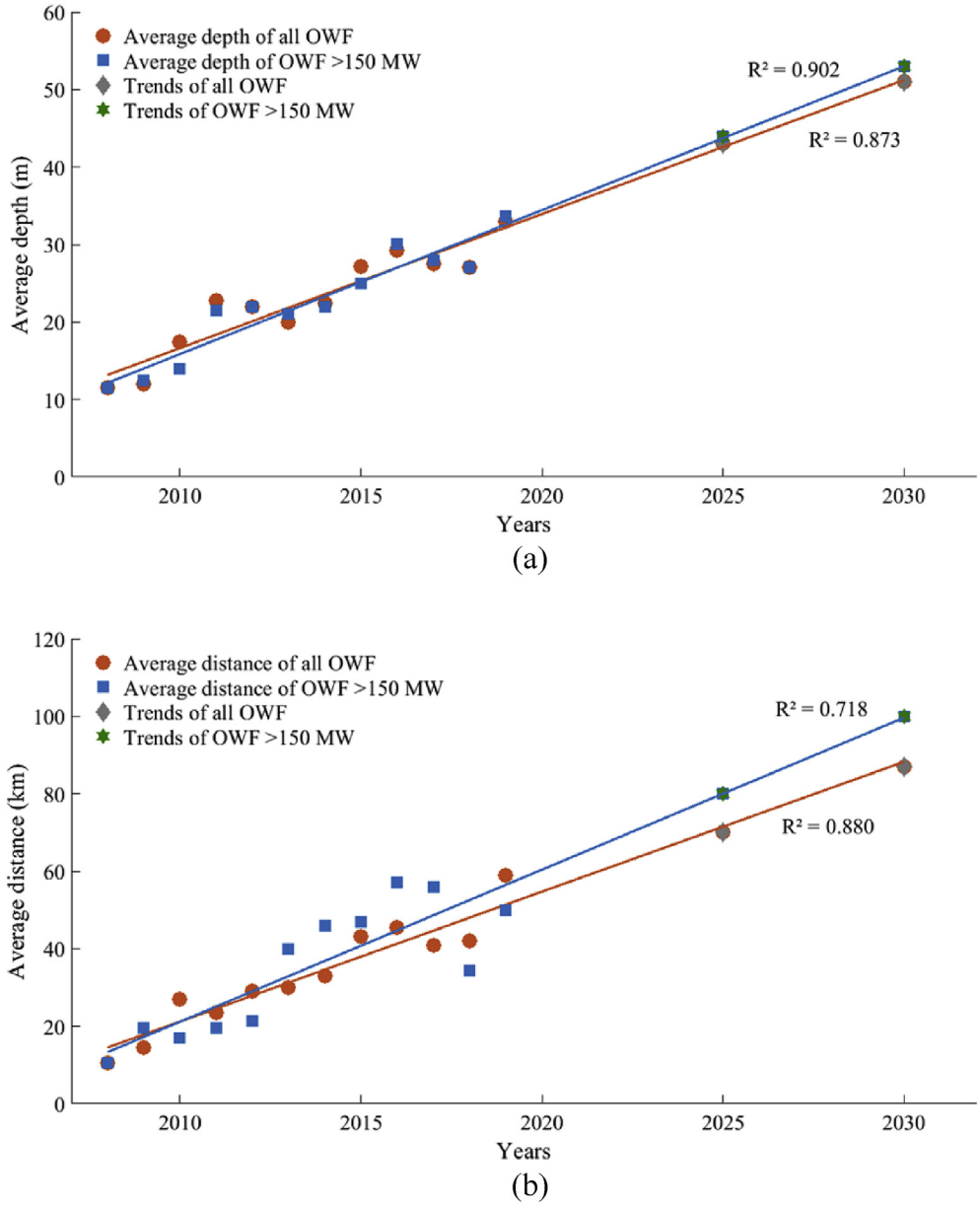

Fig. 7. Average trends for: (a) depth; and (b) distance.

[7,16].

In HVDC systems, power converters are used to adapt and control the DC voltage. The VSC-HVDC projects built up to 2010 used a two-level voltage-sourced converter technology with PWM due to its higher performance. The other alternative is the modular multilevel converter (MMC), which presents lower switching losses and a higher output voltage with low harmonic distortion [81,82].

Among the OWF observed, MVAC transmission (33 kV) was used only in the Lynn e Inner Dowsing and Westermeer wind projects. These projects have an installed capacity of 209 and 144 MW respectively. However, they are very close to the coast (6 and 1 km). As shown in Table 3, 70% of the transmission systems in OWF above 150 MW corresponds to HVAC. However, a significant increase of HVDC transmission occurred in 2013.

Currently, several plants employ HVDC transmission, and this trend increases as the distance to coast of the OWF grows. In 2015, eight OWF were connected through an HVDC link, and many more will be connected after 2018. Nowadays, Germany is the leader in OWF connected through HVDC links. Germany has twelve offshore HVDC links, while the UK has one wind farm (Global Tech 1 of 400 MW), and the next country to connect an HVDC to the coast will be Belgium.

Fig. 10 summarizes all the information in terms of the transmission technologies. The HVDC transmission was first used after 2013, and it has a total installed capacity of approximately 6 GW and an average rated power of the turbines of 5.4 MW. Despite the huge increase of HVDC, HVAC transmission still dominates the scene.

3.2.6. Array systems

Currently, there are several options for the internal topology of OWF regarding their design. The main topologies are reviewed here, and they can be divided into two groups: AC collection or DC collection [83,84].

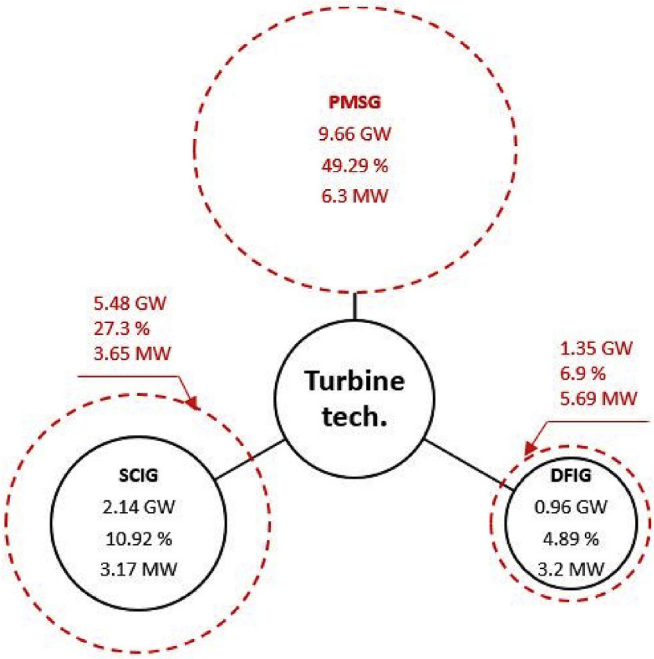

Fig. 8. Cluster diagram of turbine technologies.

AC topologies are consolidated and used in most of the cases. Radial parallel AC topology is the most common, being used in various onshore and offshore plants. However, although this topology is the cheapest and commonly applied, it is not the most adequate option for OWF. Due to this inconvenient, single-sided ring, double-sided ring, and star topologies are better options for AC collection in OWF. DC collection, or even the mix of two or more topologies, are relatively new, and currently under research. Since the array collector system can be developed from different possibilities, the best topology is sought in relation to investment vs. reliability. In this context, the aforementioned topologies are beginning to become viable options [85].

The selection between AC and DC has impacts on many aspects of the project, including the turbine technology. A voltage of 33 kV is the most frequently used in internal array systems of OWF. However, with the increase of the rated power of the turbines in OWF, the trend is to increase the voltage rating as well. Hence, a rated voltage of 66 kV has been proposed in several works, and, although it has not been implemented yet, it might be the future trend for the internal array [86–89]. Moreover, some plants authorized in Europe will use this 66 kV voltage in their internal array.

3.2.7. Foundations

There are several solutions applied to foundation structures of wind turbines installed in the sea. These solutions change according to specific project conditions, such as costs, location of installation, geotechnical conditions and depth of the OWF [27]. The first OWF was implemented in shallow waters. However, over the years, the OWF moved away from the coast into deeper waters. Different turbine foundation types are shown and described below [20,21,26,90,91], being gravity, monopile, tripod/tripile, jacket and floating the main solutions, as illustrated in Fig. 11. The black circles are divided into foundation types. The red dashed circles represent the variation of installed power for each alternative from 2013 to 2019.

The gravity foundation uses its own weight to stabilize the structure. This structure is more suitable for waters up to 30 m [21]. As shown in Table 4, by the end of 2019, 301 gravity foundations were installed in Europe.

The monopile foundation structure has been traditionally used for offshore wind turbines [21]. A simplified design procedure of monopile foundations is presented in Ref. [92]. The monopile is the most frequently used foundation in OWF in Europe, with 4258 structures installed in 2019, as observed in Table 4. The main reason for their extensive use is that many OWF are located in waters not exceeding 30 m deep [21].

The tripod/tripile structure consists of three stakes inserted from 10 to 20 m into the seabed connected to the tower through a large steel central shaft. The maximum depth suggested for these structures is 50 m, although 45 m are preferable [20,21]. It is important to emphasize that this foundation was used in the OWF Bard Offshore 1, which is located 110 km from the coast, and the results were satisfactory [20,21].

Jacket structures are composed of three or four legs trussed and inserted to the bottom of the sea through a steel tube. This structure

Table 2
Generator models installed in large OWF in Europe.

Country	Turbine Model	Technology	Number of Turbines	Total	%
GE	Bard VM 5 MW – Bard [53]	DFIG	80	562	13%
UK	Senvion 5 MW – Repower [54]		36		
BE, GE	Senvion 6.15 MW – Senvion [55]		150		
DK	V80–2.0 MW – Vestas [56]		80		
UK, BE	V90–3.0 MW – Vestas [57]		216		
GE	AD5-135-5 MW – Adwen [58]	PMSG	70	1565	36,3%
DK, GE	M5000-116-5 MW – Areva [59]		120		
GE	Haliade-150-6.0 MW – GE Renewable Energy [60]		62		
GE,NL,BE	V164–8.4 MW – Vestas [61]		180		
NL	SWT108–3.0 MW – Siemens [62]		48		
UK, GE	SWT154–6.0 MW – Siemens [63]		473		
BE, UK	V112–3.0 MW – Vestas [64]		145		
UK	V164–8.0 MW – Vestas [42]		72		
BE,UK,GE	SWT154–7.0 MW – Siemens [65]		395		
UK	SWT107–3.6 MW – Siemens [66]	SCIG	667	2180	50,7%
UK, GE	SWT120–3.6 MW – Siemens [67]		794		
GE, NL	SWT120–4.0 MW – Siemens [68]		300		
DK	SWT82–2.3 MW – Siemens [69]		253		
BE	V112–3.3 MW – Vestas [70]		50		
UK	V112–3.45 MW – Vestas [70]		116		

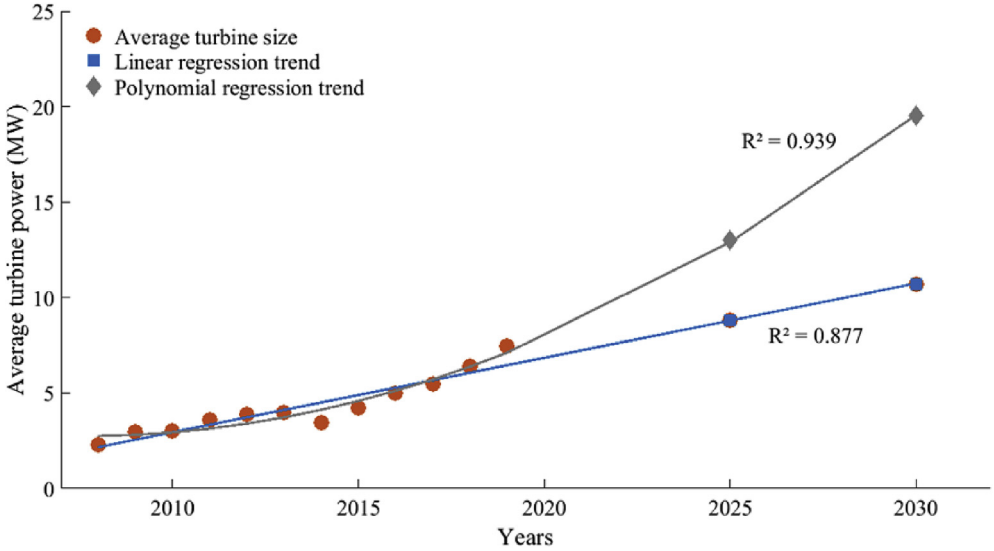

Fig. 9. Trends for average offshore wind turbine power.

Table 3
Transmission voltage level used for plants above 150 MW.

Transmission technologies		Number of Offshores	Voltage levels
HVAC	40	2	33 kV
		14	132 kV
		8	150 kV
		1	155 kV
		14	220 kV
		1	245 kV
HVDC	17	1	150 kV
		2	250 kV
		3	300 kV
		11	320 kV

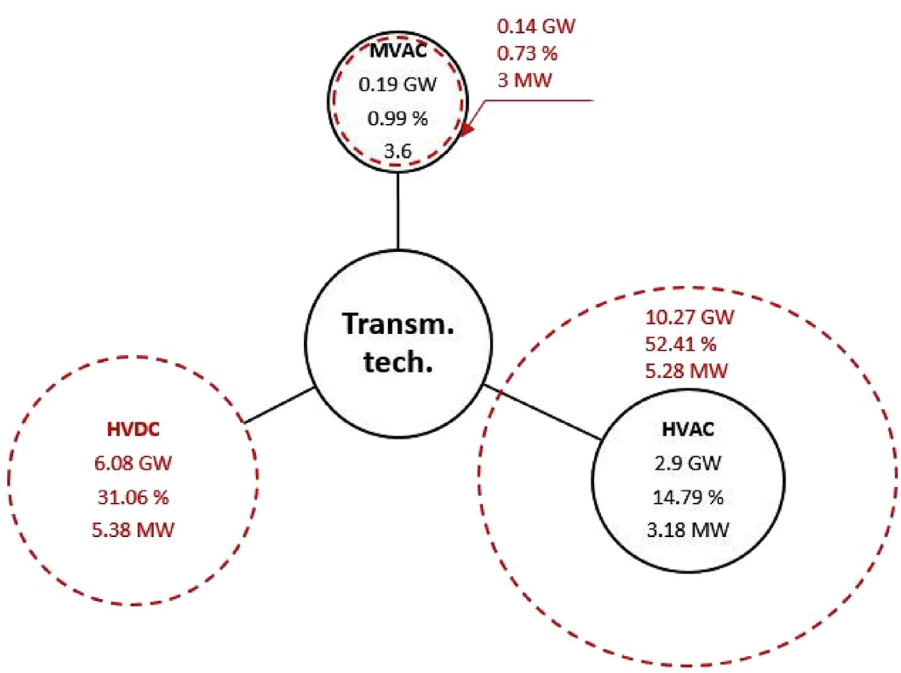

Fig. 10. Cluster diagram of transmission technologies.

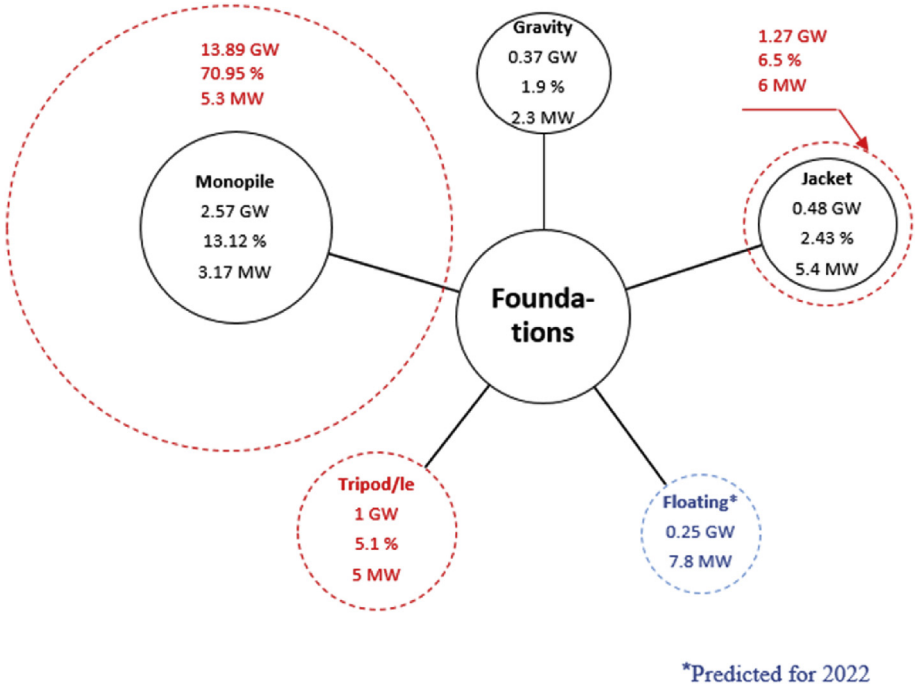

Fig. 11. Cluster diagram of foundations.

Table 4
Amount of turbine foundations installed by type between 2010 and 2019.

Year	Gravity	Monopile	Tripod/le	Floating	Jacket
2010	285	742	11	2	33
2011	286	1021	23	3	88
2012	302	1376	76	4	91
2013	303	1866	171	2	130
2014	303	2301	175	2	137
2015	303	2653	175	2	178
2016	313	3354	212	1	272
2017	283	3720	212	7	315
2018	301	4105	206	9	403
2019	301	4258	206	10	468

is used in depths that reach 80 m, although the ideal depth is no more than 45 m [20,21]. Nonetheless, some deeper water wind farms used the jacket type structure, as it can be seen in Table 4.

The floating foundation, which has an anchorage system connected to the seabed, is inspired by oil and gas platforms. The cost, construction, installation and decommissioning have proven viable for this application.

According to the analysis mentioned above for OWF in Europe, for depths up to 30 m (considered shallow waters), gravity and monopile structures are predominant. In transitional waters (from 30 to 50 m), tripod, tripile and jacket structures are used, and in deep waters (over 50 m), the floating structures are under intensive research to make these structures financially viable, although these applications are currently dominated by tripod, tripile and jacket.

3.2.8. Costs

The costs of OWF are divided into capital, variable and decommissioning costs [93]. Fig. 12 shows the composition of costs in OWF with more detail. In general, due to its greater complexity, an OWF is more expensive than an onshore with the same rated power [94]. Nevertheless, offshore energy generation begins to turn competitive with other renewable energies, and the maturity of the technology can be highlighted. Many papers present cost reductions for offshore wind generation [22,95,96]. On the other hand, the necessity to be realistic is emphasized in Ref. [97], because there are still uncertainties regarding costs.

At a global level, offshore wind power witnessed a price drop between 2010 and 2018. The global weighted average levelized cost of energy (LCOE) of offshore wind projects commissioned in 2018 was 0.11 €/kWh, which represented a decrease of 20% in relation with 2010 (0.14 €/kWh). Moreover, offshore wind projects that will be commissioned after 2020 expect to reach between 0.053 and 0.088 €/kWh [98]. The LCOE represents the average income per unit of electricity generated that would be needed to recover the costs of building and operating a generating plant during a defined financial life and duty cycle [99].

In Europe, the LCOE dropped from 0.14 to 0.12 €/kWh between 2010 and 2018, representing a decrease of 14%. Belgium reduced by 28%, Germany 24% and the UK 14% falling to 0.125, 0.11, and 0.123 €/kWh respectively. Another important observation is that the capacity factor of OWF, that is the average power generated divided by the rated peak power, increased from 38% in 2010 to 43% in 2018 [98]. In the last decade, the highest average cost of offshore power occurred in 2014, reaching 4.77 M€/MW. This is justified by the fact that, between 2012 and 2015, OWF reached deeper waters, and the challenges faced were enormous. Moreover, in 2018, offshore average costs reached 3.87 M€/MW, showing a reduction of approximately 20% compared to 2014.

Fig. 13 presents the global weighted average of LCOE and capacity factor for different renewable energy systems. It can be observed that, although OWF have a slightly higher price per kWh when compared to onshore wind farms and solar photovoltaic, their LCOE has been decreasing over the last years. Moreover, the capacity factor of OWF is increasing with the higher rated power of the turbines.

Due to the tendency of OWF to move away from the coast, projects with HVDC are becoming more common. When analyzing the costs of the OWF above 150 MW, it is verified that the HVDC

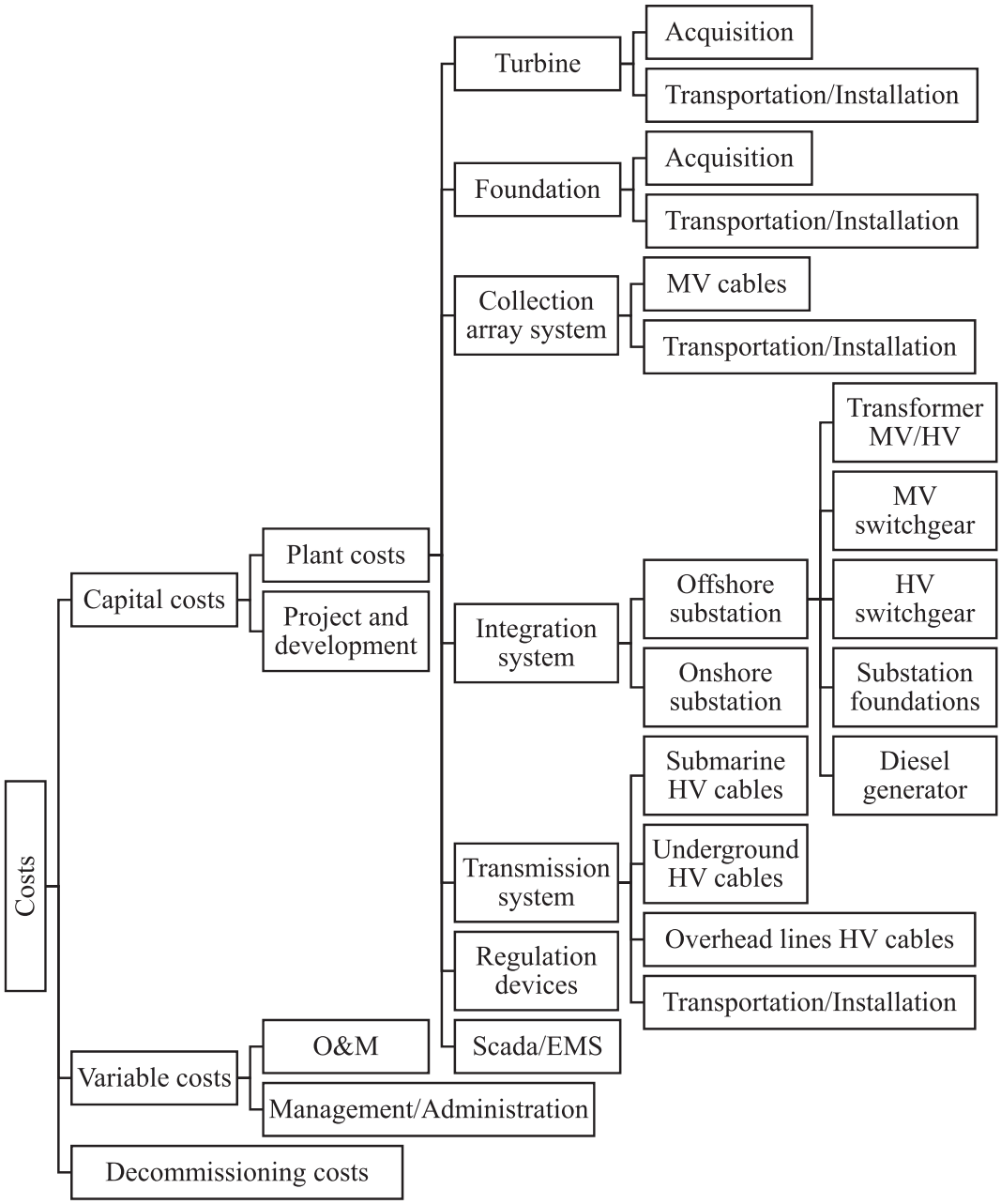

Fig. 12. Distribution of costs for OWF.

technology is currently more expensive than the HVAC, which has matured the most in the recent years. However, from the first plant in HVDC (Bard Offshore 1), its cost has been decreasing gradually. Other illustrative examples are the first phases of Trianel Borkum 1 and Borkum Riffgrund 1, whose costs are 20% and 38% higher than their second phases respectively, even though they present the same characteristics.

For AC technology, there is a slight increase in costs per MW. Hence, when the first phases of Belwind 1 and Horns Rev 1 are analyzed, their costs are approximately 10% and 20% lower than their second phases respectively.

4. Large-scale OWF by country

Among the five countries operating large offshore, Germany and the UK are the world leaders. However, the power profile of these countries is completely different. It is important to emphasize that other countries, such as Finland, Ireland, Norway, Sweden, Spain, and France, are also responsible for the offshore generation, but none of them has farms above 150 MW. Additionally, a review of other countries outside Europe has been carried out, which revealed that only China presents relevant offshore generation with large-scale OWF.

4.1. Germany

Germany currently contributes with about 7105 MW of OWF above 150 MW, which corresponds to 36% of the large offshore wind power in Europe. In addition, it has more than 1400 wind turbines installed in this segment. Germany installed around 29% of

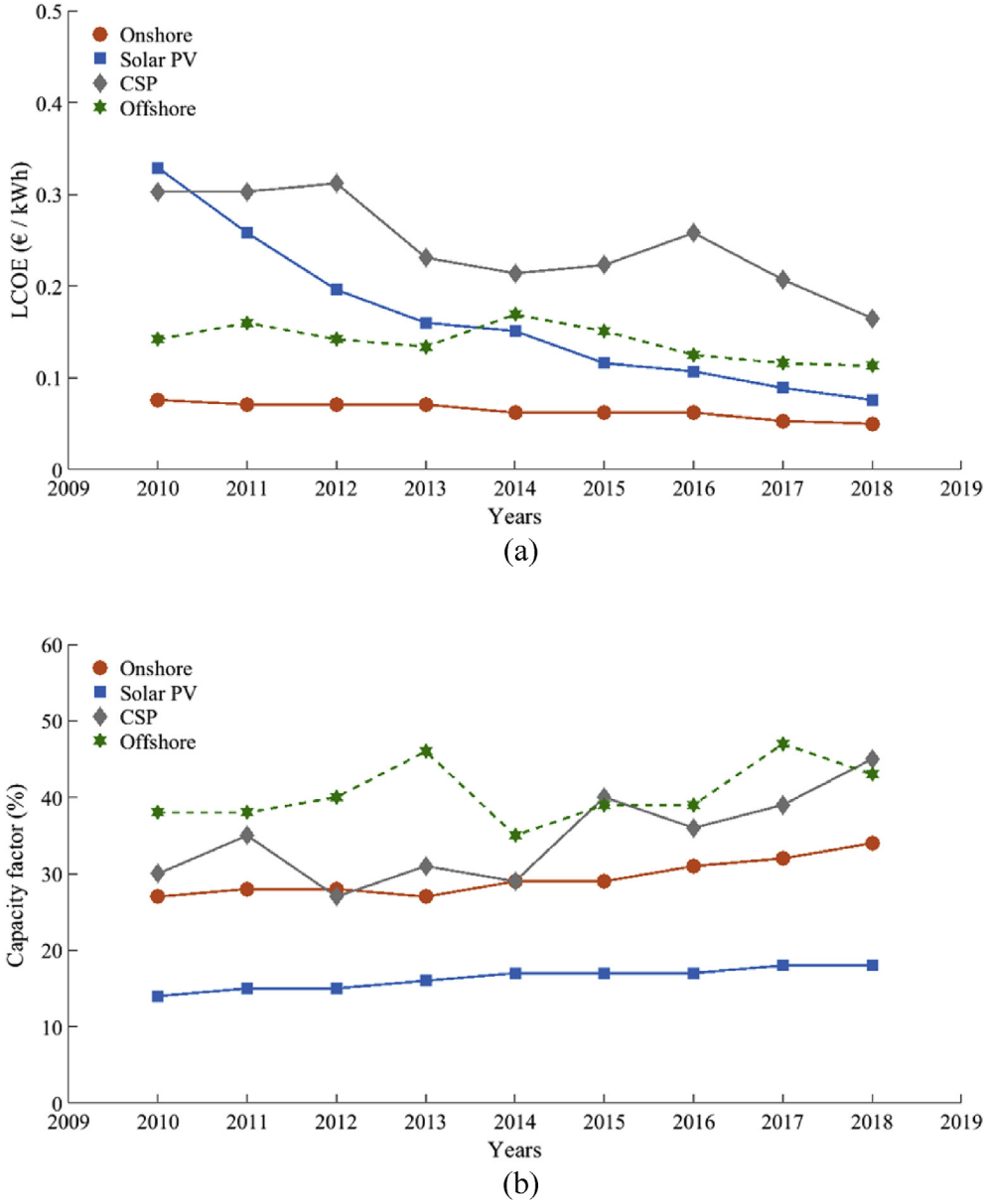

Fig. 13. (a) Global weighted average of LCOE; and (b) capacity factors for renewable systems.

all new offshore capacity in 2017, with 1.3 GW [38], and pursues to reach 15 GW by 2030.

Everything indicates that offshore energy in Germany will continue to grow, given the incentives put forward. For example, the KfW Bankengruppe is a German government-owned development bank funding up to ten OWF. Despite several existing incentives, Germany demonstrates that offshore will be even more competitive in the coming years, since two projects to be commissioned in 2024, and one in 2025, won bids with values that are competitive against market rates without subsidies [44].

Regarding the characteristics of OWF in Germany, the average depth of the wind turbines is between 25 and 30 m, while the average distance to the coast is between 40 and 60 km (eight out of the twenty OWF are located between these distances). The deepest wind turbine is founded at 44 m, while the furthest OWF are Veja Mate and Bard Offshore 1, located 114 and 112 km away from the coast respectively.

In terms of the offshore wind turbine technology, about 47% correspond to SCIG with full converter, followed by a 40% of PMSG, and a 13% of DFIG. In 2018 and 2019, almost 300 new PMSG-driven turbines were installed, being the dominant trend in this country.

The HVDC transmission system is preferred in Germany. Among the twenty plants analyzed, sixteen of them use HVDC transmission, being 320 kV the typical voltage rating. This choice is justified by the long distance to the coast, since this technology becomes more viable as the distance to the coast increases. As for the array systems, all the OWF in Germany present voltage ratings between 30 and 34 kV. This trend will continue in future projects also.

The main foundation used in Germany is the monopile. However, Germany uses more tripod and jacket foundation compared to other countries. This is mainly due to the high average depth of the turbines installed.

4.2. United Kingdom

Since the government established the first incentives for the development of OWF in the year 2000, the UK has experienced a relevant growth of this segment. In 2014, the UK was the largest investor in renewable energy in Europe. Projects in the offshore sector accounted for 86% of the wind energy investments. For example, these resources were destined for the Dudgeon and West of Duddon Sands plants, with 402 MW and 389 MW respectively [9,44]. Moreover, the UK has ambitious offshore deployment plans and wants to reach 16 GW of offshore installed capacity by 2020, and about 40 GW by 2030 [9,100,101].

Currently, the UK is the country that contributes the most to large OWF. In 2019, the UK had about 8633 MW installed, which corresponds to 44% of the total large offshore capacity in Europe. In addition, there were 1916 wind turbines installed in OWF above 150 MW as of 2019.

The average depth of the wind turbines in UK is between 18 and 25 m, while the average distance to the coast is between 20 and 25 km. However, thirteen out of the twenty-three plants are located more than 15 km away from the coast. The deepest wind turbine is founded at 56 m, while the furthest OWF is about 120 km away from the coast.

Regarding the wind turbine technology, SCIG with full converter was the most common choice in 2019, with 56% of the total, followed by PMSG with a 34%, and DFIG with a 10%. Like in Germany, PMSG begins to be the trend for offshore power generation in this country, since 660 PMSG have been installed after 2015 in large OWF, against 276 SCIG with full converter.

The transmission system mostly used in the UK is HVAC at 132 kV and 220 kV. This choice is justified by the reduced distance to the coast. At short distances, this technology is more viable that HVDC. Regarding the array systems, all the OWF present a rated voltage between 33 and 36 kV. However, this trend may change in future projects, where 66 kV array systems will be used. Fig. 14(a) illustrates the main characteristics of OWF installed in Germany and (b) in the UK.

4.3. Belgium, Denmark and the Netherlands

Belgium, Denmark and the Netherlands contribute with 20% of the offshore generation in Europe. There are twenty-five OWF distributed among these countries.

The Belgian government is encouraging the production of offshore energy and has allowed an area of 240 km^2 for offshore projects with potential to generate approximately 2250 MW. This area is expected to be expanded after 2020 [7]. Currently, Belgium has six OWF farms above 150 MW connected to grid, corresponding to 1556 MW of power through 318 turbines. Moreover, the government has confirmed adding about 720 MW before the end of 2020 [26].

The growth in size and power of the OWF in Belgium can be observed in the fact that six out of the seven OWF in this country are above 150 MW. Besides, all of them were built after 2013, reinforcing the trend to larger OWF over the years. The average depth and distance to shore are around 25 m and 35 km respectively. In addition, the furthest OWF is 45 km away from the coast, and the deepest is founded at 38 m. The main transmission technology in Belgium is HVAC with array systems at 33 kV. The preferred generator is the PMSG, with 50% of the turbines installing this technology.

Denmark has fourteen OWF connected to the grid, summing an installed capacity of 1703 MW distributed among 559 turbines. The tendency observed for Denmark is also towards larg projects. Currently, Denmark has six OWF that contribute with 1549 MW. Like Belgium, Denmark has also extended its offshore potential, and intends to increase its installed power in 1357 MW by 2021, spread among Horns Rev III (400 MW), already full commissioned, Nearshore 2 (350 MW) and Kriegers Flak (600 MW). Denmark pursues a new world record in the price of offshore wind with these OWF.

The average depth and distance to shore of OWF in Denmark are around 16 m and 36 km respectively, being Global Tech I the furthest and deepest (114 km away from the coast and founded at 41 m). The main transmission technology is HVAC with array systems at 33 kV, and 55% of the installed turbines are based on SCIG with full converter.

The Netherlands contribute with 744 MW and 198 turbines to the European offshore power generation. The Dutch government incentives offshore wind projects, and aims for an installed capacity of 4.5 GW by 2023, and a subsequent construction of 1 GW/year thereafter until 2030 [31]. The deepest and furthest OWF are around 36 m and 32 km respectively. The main transmission technology is HVAC at 220 kV, with array systems at 33 kV. Furthermore, turbines equipped with SCIG with full converter are typically used. The main characteristics of OWF, compared by country, are shown in Table 5. It can be observed that the UK presents the highest installed capacity and number of turbines, whereas Germany is the leader in HVDC transmission, and Denmark has the most distant OWF and the lowest average cost of CAPEX.

4.4. OWF outside Europe

Although Europe is the main contributor when it comes to OWF, it is possible to find some power plants above 150 MW operating in China. China is the third producer of offshore wind energy in the world with an installed capacity of 4.3 GW, and other 4 GW of OWF under construction [102], whereas the target is to reach 5 GW by 2020 [103]. When analysing only plants above 150 MW, Fig. 15 shows the installed and accumulated capacity, where a significant growth can be noticed. The construction of OWF began with a single plant in 2010, and it started growing in 2016. Other upcoming markets that have shown interest in OWF are Japan, Korea, Taiwan, USA, India and Vietnam, but in these countries the OWF are not in operation, nor classified as large-scale plants.

Fig. 16 shows the cumulative capacity of OWF in MW and the projections for 2031 according to Ref. [102]. It can be observed that the estimation in installed capacity for China is approximately 39 GW for 2031, which is below the 49 GW estimated for Europe.

In terms of water depth, the early commissioned projects in China came from nearshore plants, which include depths ranging from 10 to 15 m. Previously, the water depth was mainly less than 10 m [102,104]. When compared to Europe, it can be noticed that the Chinese OWF are located in shallow waters, while the European plants are in deeper waters, being the average depth 33 m. Regarding the distance to shore, the European OWF are furthest from the coast (50 km average) than the Chinese plants (25 km average) [36,102].

In China there are approximately 1100 wind turbines installed offshore, as it can be observed in Table 6, while in Europe there are about 4307 wind turbines for plants above 150 MW (see Table 2). The average rated power of the wind turbines used in the European OWF nowadays is approximately 7.5 MW, whereas it decreases to 4.1 MW in the Chinese OWF [36]. The European OWF already use the HVDC technology as transmission system (see Table 3), while only HVAC technology, mainly with a rated voltage of 220 kV, is used in the Chinese OWF [36]. For the internal array system, the voltage of 33 kV is the most frequently used in the OWF in Europe, while China typically uses a standard of 35 kV.

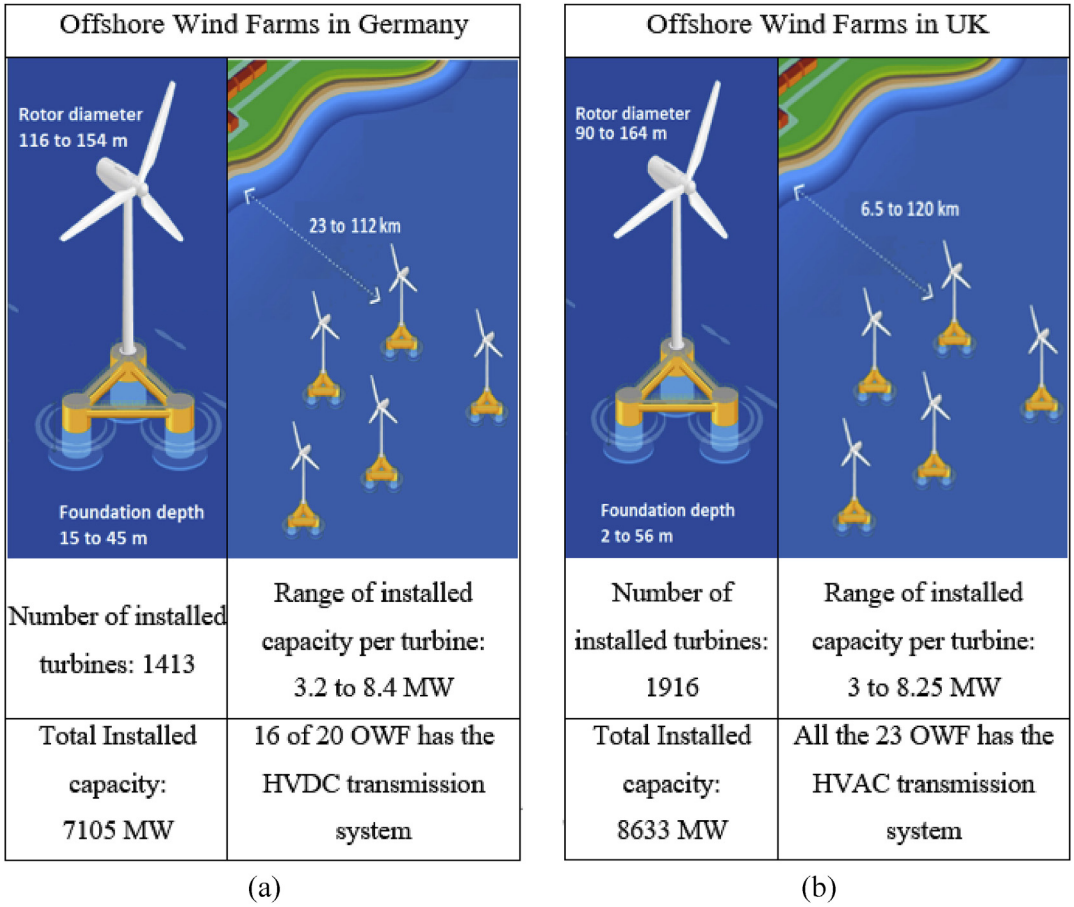

Fig. 14. Difference between the profile of OWF in: (a) Germany; and (b) the United Kingdom.

Table 5
Comparision between OWF by country.

Country	Capacity (MW)	Number of turbines	Turbine rated power (MW)	Predominant turbine model	Depth (m)	Distance to shore (km)	Main transmission type	Cost (M€/MW)
UK	8633	1916	3–8.4	SCIG	2–56	6.5–120	HVAC 220 kV	4.14
GE	7105	1413	3.6–8.4	SCIG	15–45	23–112	HVDC 320 kV	4.25
BE	1556	318	3–8.4	PMSG	13–38	28–46	HVAC 220 kV	3.95
DE	1549	462	2–8.4	SCIG	6–41	9–114	HVAC 150 kV	2.71
NL	744	198	3–4	SCIG	5–36	5–32	HVAC 220 kV	3.77

5. Environmental issues and constrained potential

The OWF belong in the green industry context, and, in this sense, the intention is to maintain the green label associated with them by not causing and/or reducing the damage to the environment when manufacturing or installing these plants [105]. However, as in any human activity, a certain impact on the surrounding ecosystem is inevitable. According to Ref. [27], there are positive and negative issues associated with OWF. The main positive effects are the replacement of fossil fuels with wind energy, and the offshore structures acting as artificial reefs, increasing the diversity of marine species. In contrast, the risks of collision for birds, the displacement of marine mammals and birds, and the habitat loss or degradation, can be mentioned as negative impacts. Some measures during the planning process can reduce the negative impacts associated with OWF. For instance, the authors in Ref. [106] mention that the proper election of an OWF location includes choosing a place outside protected areas and scheduling the construction operations outside the breeding season.

Regarding the legal framework, a proper regulation is required in view of the remarkable expansion of the offshore plants and in order to avoid harmful impacts. To deal with the legal aspects of OWF, the Environmental Impact Assessment (EIA) is essential. Through the EIA, the impacts of the project are identified and measures are adopted to mitigate these adverse effects. Any OWF with potential to cause adverse impact must be able to produce an appropriate EIA, and if significant disturbances of the surrounding environment are verified, the power plant construction can only continue with compensatory measures [107].

Following the same reasoning towards a sustainable development, there is an important legislative structure at European level, such as the Birds and Habitats Directives (79/409/EEC and 92/43/EEC, respectively), which provide a structure for preserving species and habitats of interest, including the designation of Special Protection Areas (SPAs) and Special Areas of Conservation (SACs) under the Natura 2000 network [27]. Natura 2000 is a network of

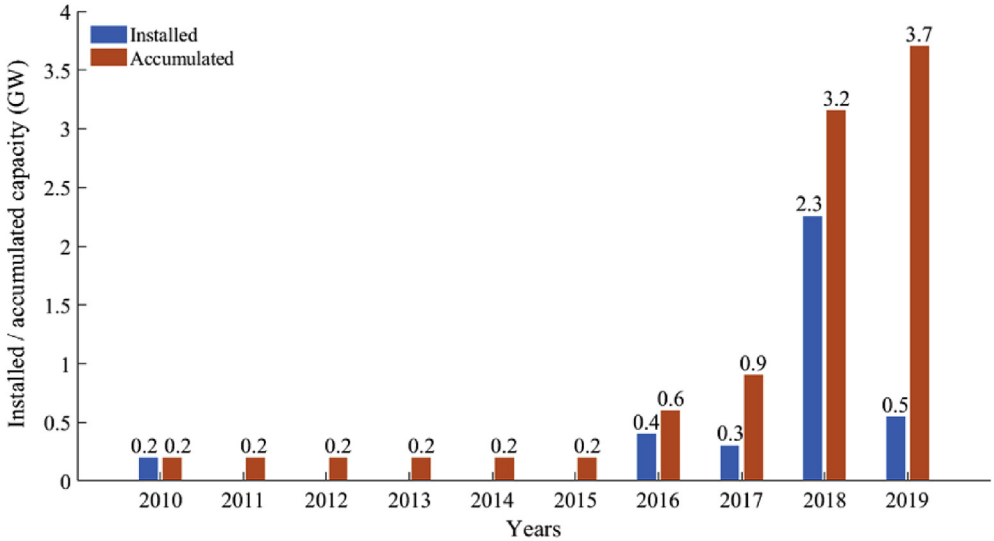

Fig. 15. Installed capacity of the largest OWF installed in China [36].

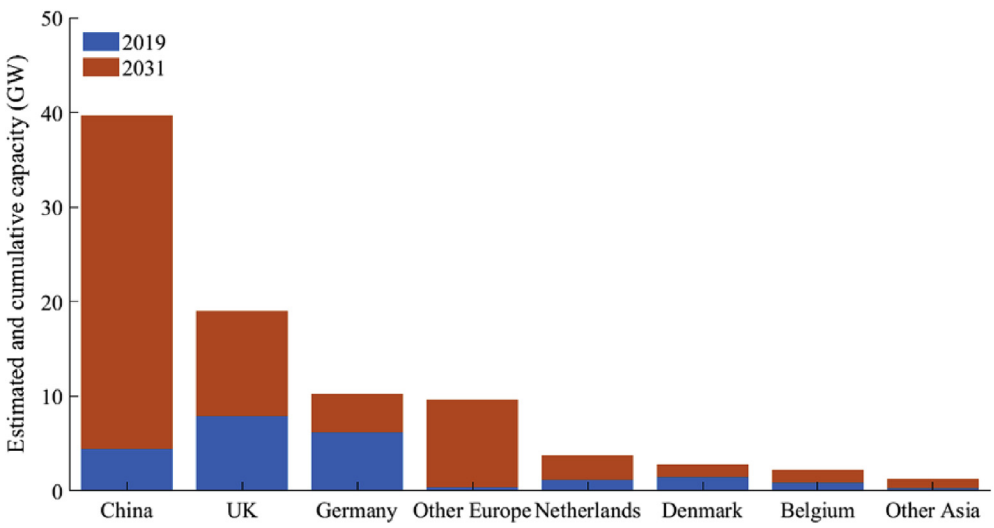

Fig. 16. Estimated 2031 cumulative offshore wind capacity by country by country [102].

Table 6
Average parameters of OWF in China above 150 MW [36].

Total Nº WTs	Avg. Rating (MW)	Avg. Capacity (MW)	Avg. Depth	Avg. Dist. to shore (km)	Avg. Array (kV)	Avg. Transm.
1082	4	265	10	26	35	220 kV (HVAC)

protected areas on land and sea across the 28 countries of the European Union (EU) for rare and threatened species and habitats [108]. If a certain location is identified as sensitive, this place should not be used for the construction of OWF, especially under the presence of endangered species. The Convention on the Conservation of Migratory Species of Wild Animals (Bonn Convention) on wildlife protection also grants responsibilities on signatory parties. This convention provides a platform for the conservation of migratory birds, aquatic or not, and their habitats. A Marine Strategy Framework Directive (MSFD) is working on marine management strategies to achieve a beneficial environmental status by 2021. This directive is part of the EU legislation specifically targeted at the protection of the marine environment and natural resources for a sustainable use of the waters [39]. More detailed information regarding the legal framework and obligations in Europe can be found in Ref. [107].

The offshore plants have to coexist in harmony with shipping routes, military use, oil and gas extraction, natural protected areas and tourist zones, due to the varied economic exploitation of the seas. Therefore, the potential for OWF development is limited. Currently, only 4% of the area with a distance to shore between 0 and 10 km can be used for OWF applications. This figure increases to 10% for distances between 30 and 50 km, and 25% for distances above 50 km [39]. Another important issue open to debate is the visual impact caused by OWF. This impact is considerable for OWF less than 10 km away from the coast. For this reason, the

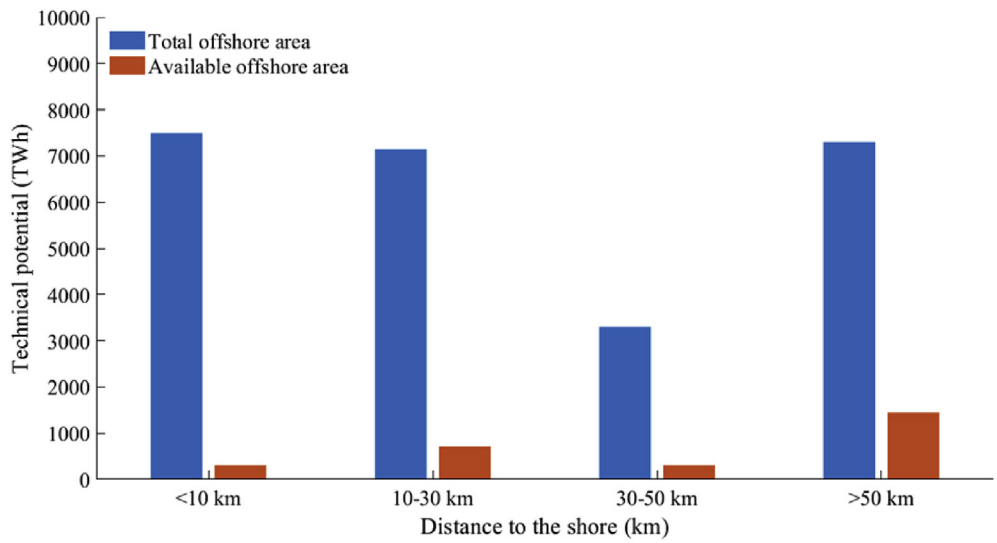

Fig. 17. Available potential for OWF [39].

Netherlands and the UK have prohibited building OWF less than 22 km away from the coast [27,39]. Fig. 17 illustrates the total and the available unrestricted technical potential in TWh in the European seas. According to Ref. [39], if there were no restrictions, the amount of electricity generated by OWF would be enough to supply about 60% of the energy demand in Europe by 2030.

The aforementioned points highlight the difficulty to find the balance between positive and negative impacts. Therefore, it becomes necessary to reach a trade-off between a sustainable development and an ever-growing electricity demand.

6. Conclusions

In this work, future trends have been pointed out from the analysis of the information gathered. The main conclusions are summarized below:

- The installed power estimation for 2025 and 2030 is 47.4 and 76 GW, respectively. The estimated number of OWF for 2025 and 2030 is 139 and 172.
- Europe is the main participant in terms of OWF above 150 MW (the UK and Germany lead with the 80% of the installed capacity). Nevertheless, China is expanding its contribution.
- The new OWF are moving away from the coast into deeper waters. For OWF above 150 MW, the average depth predicted for 2025 and 2030 is 44 and 53 m; and the distance estimations are 80 and 100 km, respectively.
- The main foundation technology is the monopile, used in 80% of the cases. However, there is a tendency towards the jacket due to the increase in the turbine power and depth.
- The polynomial regression for the offshore wind turbine average rated power estimates 12.7 and 18.8 MW for 2025 and 2030; whereas the linear regression indicates 8.6 MW and 10.7 MW, respectively.
- There is an increase in the use of HVDC transmission lines justified by the larger distances to the coast and decreasing costs of the technology.
- There is a tendency towards 66 kV in the internal collection systems of the OWF as the rated power of the wind turbines grows. In addition, due to the increase in HVDC transmission, some new internal configuration topologies in DC are being proposed.
- Offshore energy costs are competitive compared to other renewable sources. The global weighted average LCOE decreased 13% in the period 2010–2017. Projects to be commissioned after 2020 expect a 22% reduction of CAPEX compared to the period 2013–2017. OPEX costs can decrease between 27% and 43% by 2025.

The present study gathers the most representative characteristics of large OWF in Europe. The figures given envisage a relevant growth of this type of power source for the following years. The development and increasing use of the OWF rely on a reduction in costs and a further maturation of the technology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Emanuel P.P. Soares-Ramos: Conceptualization, Formal analysis, Investigation, Methodology, Writing - original draft. **Lais de Oliveira-Assis:** Conceptualization, Formal analysis, Investigation, Methodology, Writing - original draft. **Raúl Sarrias-Mena:** Conceptualization, Formal analysis, Methodology, Supervision, Writing - review & editing. **Luis M. Fernández-Ramírez:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing - review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2020.117787.

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